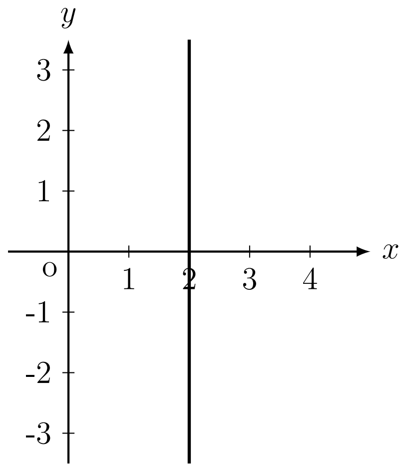
MSET004

Question 1.



[**Tg:@abusat_760**] Which of the following is an equation of the line shown in the xy-plane above?

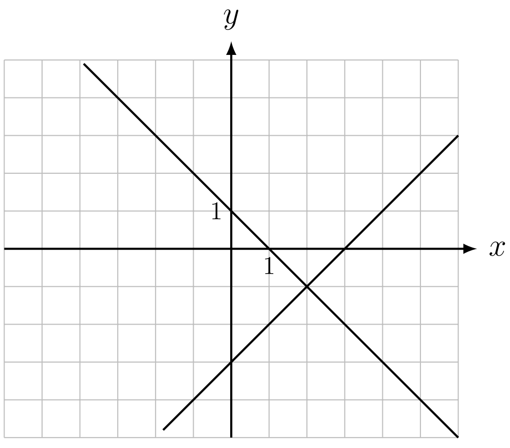
A) $y = 2$

B) $x = 2$

C) $y = 2x$

D) $x = 2y$

Question 2.



[**Tg:@abusat_760**] A system of two linear equations is graphed on the xy-plane above. Which of following ordered pairs (x, y) is the solution to the system?

A) $(1, 1)$

B) $(1, -2)$

C) $(2, -1)$

D) $(2, 1)$

Question 3.
[**Tg:@abusat_760**] Salma ordered a set of golf clubs and a golf bag online, which were shipped to her house. The weight w, in pounds, of the bag and the clubs is estimated by the equation $w = 1.5c + 12$, where c is the number of clubs in the bag. What is the best interpretation of the number 1.5 in the equation?

A) The estimated weight, in pounds, of 1 club

B) The estimated weight, in pounds, of 12 clubs

C) The estimated weight, in pounds, of the bag with no clubs

D) The estimated weight, in pounds, of the bag with 12 clubs

Question 4.
[**Tg:@abusat_760**] In the xy-plane, what is the slope of the line that passes through the points $(0, 0)$ and $(4, 3)$?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) 3

D) 4

Question 5.
[**Tg:@abusat_760**] A scooter rental company charges \$9 for the first hour of renting a scooter and \$5 for each additional hour. Which of the following functions gives the cost $s(t)$, in dollars, of renting a scooter for t hours?

A) $S(t) = 9t$

B) $S(t) = 9t + 6$

C) $S(t) = 5t + 9$

D) $S(t) = 5t + 4$

Question 6.

Hours (x)	Total cost (y)
1	9
3	14
7	24
9	29

[Tg:@abusat_760] Selected values representing the total cost y, in dollars, to rent a bike for x hours are shown in the table above. The relationship between x and y is linear. If the data are graphed in the xy-plane, what is the slope of the line that represents this situation?

A) $\frac{1}{2}$

B) 1

C) 2

D) $2\frac{1}{2}$

Question 7.
[Tg:@abusat_760] In the xy-plane, line M passes through the point $(2, 4)$ and has slope $\frac{3}{2}$. Which of the following is an equation of line M?

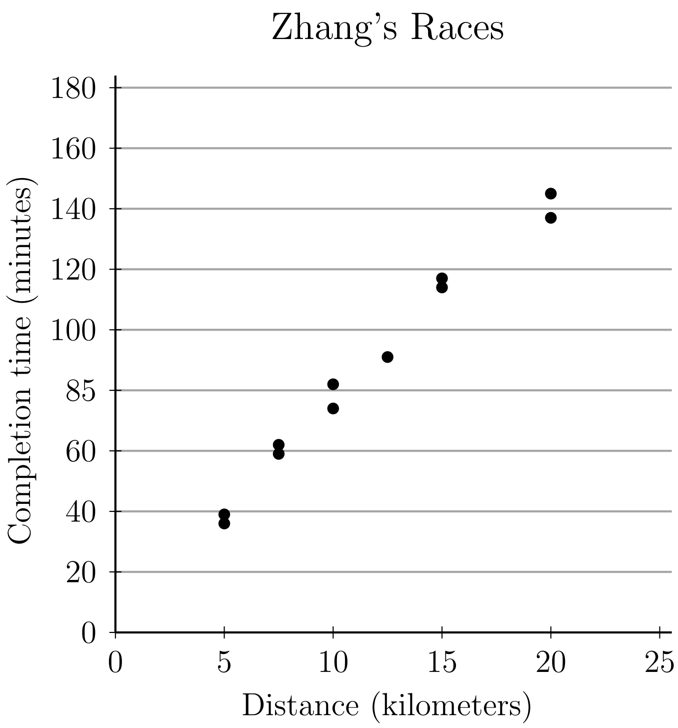
A) $y = \frac{3}{2}x + 4$

B) $y = \frac{3}{2}x + 1$

C) $y = \frac{3}{2}x - \frac{7}{2}$

D) $y = 2x + \frac{3}{2}$

Question 8.
[Tg:@abusat_760] Zhang has run 11 races this year. The distance of each race and the number of minutes it took him to complete each race is shown in the scatterplot below.



Which of the following is closest to the slope of a line of best fit for this scatterplot?

A) 2

B) 4

C) 8

D) 11

Question 9.
[Tg:@abusat_760] In the xy-plane, what is the x-intercept of the line that has a slope of $\frac{4}{3}$ and passes through the point $(0, 8)$?

A) -8

B) -6

C) 3

D) 8

Question 10.

$$h = 70 + 5m$$

[Tg:@abusat_760] A runner has a resting heart rate of 70 beats per minute. For every minute he runs, his heart rate increases by a constant number of beats per minute. The runner's heart rate, h, is modeled by the equation above as a function of m minutes of running. Based on this equation, what is the increase in heart rate for every five minutes of running?

Question 11.
[**Tg:@abusat_760**] Which of the following is the equation of a line with x-intercept $(6, 0)$ and y-intercept $(0, 15)$?

A) $y = \frac{5}{2}x - 15$

B) $y = -\frac{5}{2}x - 15$

C) $y = \frac{5}{2}x + 15$

D) $y = -\frac{5}{2}x + 15$

Question 12.
[**Tg:@abusat_760**] The amount of water y, in gallons, in a reservoir x minutes after the reservoir begins to drain can be modeled by the equation $y = 1,200 - 4x$. If the equation is graphed in the xy-plane, which of the following is the best interpretation of the x-intercept?

A) The reservoir drains at the rate of 4 gallons per minute.

B) There are 300 gallons of water in the reservoir before it begins to drain.

C) There are 1,200 gallons of water in the reservoir before it begins to drain.

D) The water will completely drain from the reservoir 300 minutes after the reservoir begins to drain.

Question 13.

$$C(x) = 7000 + 14x$$

[**Tg:@abusat_760**] A company uses the function above to estimate the cost of $C(x)$, in dollars, to produce x units of a product. Based on the model, how many units of the product can the company produce at a cost of $35,000?

Question 14.
[**Tg:@abusat_760**] What is the equation of the line that passes through the point $(8, -3)$ and is perpendicular to the line $2x - 3y - 10 = 0$?

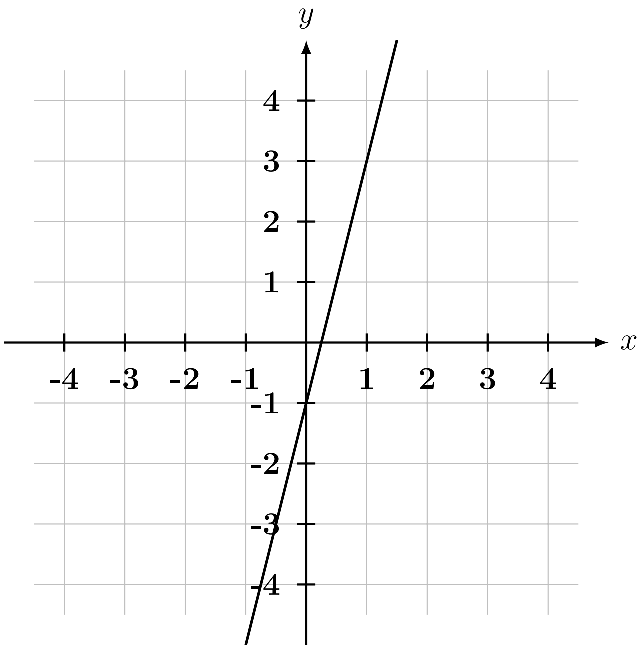
A) $y = -\frac{3}{2}x + 9$

B) $y = -\frac{3}{2}x + \frac{7}{3}$

C) $y = -\frac{3}{2}x - 15$

D) $y = -\frac{1}{2}x + 1$

Question 15.



[**Tg:@abusat_760**] The graph of a line is shown in the xy-plane above. Which of the following is an equation of the line?

A) $y = \frac{1}{4}x - 1$

B) $y = x - 1$

C) $y = 4x - 1$

D) $y = 4x + 1$

Question 16.

$$T(d) = 3.5d + 16$$

[**Tg:@abusat_760**] The function T defined above models the typing speed, in words per minute, of a student taking a typing class, where d is the number of days the student has been in class and $0 \leq d \leq 12$. According to the model, what is the typing speed, in words per minute, of a student who has been in class for 5 days?

A) 19.5

B) 30.0

C) 33.5

D) 37.0

Question 17.
[Tg:@abusat_760] The graph of $y = g(x)$ is a line in the xy-plane that has a slope $\frac{3}{2}$, if $g(8) = 1$, which of the following functions could represent $g(x)$?

A) $g(x) = \frac{-3}{2}x$

B) $g(x) = \frac{3}{2}x - 11$

C) $g(x) = \frac{3}{2}x - 9$

D) $g(x) = \frac{3}{2}x - \frac{3}{2}$

Question 18.
[Tg:@abusat_760] What is the slope of the line m with equation $2x - 5y = 10$?

A) $-\frac{2}{5}$

B) $\frac{5}{2}$

C) $\frac{2}{5}$

D) -2

Question 19.
[Tg:@abusat_760] A service delivers packages each day at an average rate 12 packages per hour. Which of the following best approximates the number of packages, n, the service delivers in t hours?

A) $n = t + 12$

B) $n = t - 12$

C) $n = 12t$

D) $n = \frac{t}{12}$

Question 20.
[**Tg:@abusat_760**] Which of the following equations, when graphed in the xy-plane, would result in a line with slope of 3 that passes through the point $(0, -4)$?

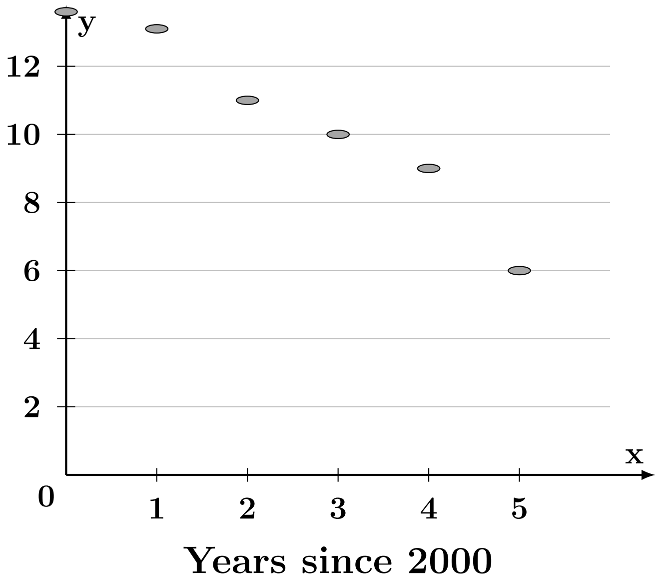
A) $y = -4x + 3$

B) $y = -3x + 4$

C) $y = 4x - 3$

D) $y = 3x - 4$

Question 21.
[**Tg:@abusat_760**] The population Y, in tens of thousands, of a certain city X year since 2000 is shown in the scatterplot below.



If the data are modeled using a line of best fit, which of the following could be an equation of that line?

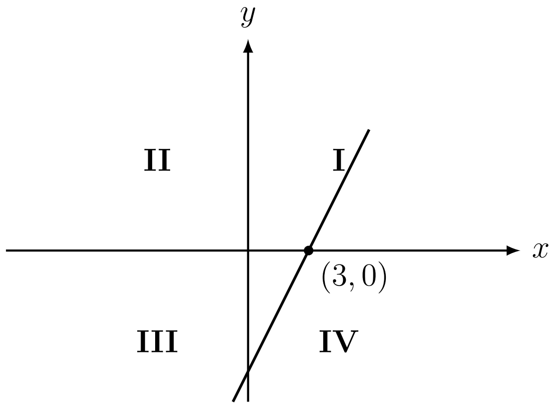
A) $y = 13.6 - 1.35x$

B) $y = 13.6 + 1.35x$

C) $y = 12.1 + 1.35x$

D) $y = 12.1 - 1.35x$

Question 22.



[**Tg:@abusat_760**] The graph in the xy-plane above shows the linear function f. if $g(x) = -4x + 8$, in which quadrant does the graph of $y = f(x)$ intersect the graph of $y = g(x)$?

A) I

B) II

C) III

D) IV

Question 23.
[**Tg:@abusat_760**] A school district currently has 2000 students and expects growth for the next 5 years. The school board is comparing two different growth projections. One model predicts that each year there will be 30 more students than the previous year. The other model predicts that each year there will be 1% more students than the previous year. To the nearest whole number, what is the difference in the projection of the total number of students these models predict after 5 years?

A) 10

B) 20

C) 29

D) 48

Question 24.
[**Tg:@abusat_760**] Which of the following is TRUE about the line whose equation is $4x - 2y - 10 = 0$?

A) The x-intercept is 4 and the y-intercept is -2.

B) The x-intercept is $\frac{5}{2}$ and the y-intercept is 5.

C) The x-intercept is $\frac{5}{2}$ and the y-intercept is -5.

D) The x-intercept is 5 and the y-intercept is $\frac{5}{2}$

Question 25.
[**Tg:@abusat_760**] Angel works part-time as a model for \$50 an hour, and part-time as a pastry chef for \$12.50 an hour. This past week, he worked for 25 hours and made \$625. How many hours did he work as a chef?

Question 26.
[**Tg:@abusat_760**] Where are the points $(1, 2)$ and $(-1, 1)$ in relation to the line $3x + 4y = 7$?

A) Both are on the line...

B) One is on the line and the other is off the line.

C) Both are above the line.

D) One is above the line and the other is below the line.

Question 27.
[**Tg:@abusat_760**] Logan runs x miles per day. Which of the following represents the total distance he runs in a year, if he takes one day off per week, and a week off every 3 months?

A) $\left(52 - \frac{12}{3}\right)(6)x$

B) $\left(52 - \frac{52}{3}\right)(6)x$

C) $\left[(365)\left(\frac{6}{7}\right) - \frac{12}{3}\right]x$

D) $\left[(365)\left(\frac{6}{7}\right) - 21\right]x$

Question 28.
[**Tg:@abusat_760**] Which of the following statements is true about the graph of the equation $4y + 5x = 7$ in the xy-plane?

A) it has a negative slope and a positive y-intercept

B) it has a negative slope and a negative y-intercept

C) it has a positive slope and a positive y-intercept

D) it has a positive slope and a negative y-intercept

Question 29.
[Tg:@abusat_760] The front of a roller-coaster car is at the bottom hill and is 20 feet above the ground. If the front of rollercoaster car rises at a constant rate of 6 feet per second, which of the following equations given height h in feet, of the front of the roller-roller coaster s seconds after it starts up the hill?

A) $h = 6s + 20$

B) $h = 20s + \dfrac{120}{6}$

C) $h = 6s + \dfrac{120}{6}$

D) $h = 20s + 6$

Question 30.

$$c = 25h + 50$$

[Tg:@abusat_760] The equation above gives the amount C, in dollars, an electrician charges for a job that takes h hours. Ms. Ghada and Mr. Amr each hired this electrician. The electrician worked 3 hours longer on Ms. Ghada's job than on Mr. Amr's job. How much more did the electrician charge Ms. Ghada than Mr. Amr?

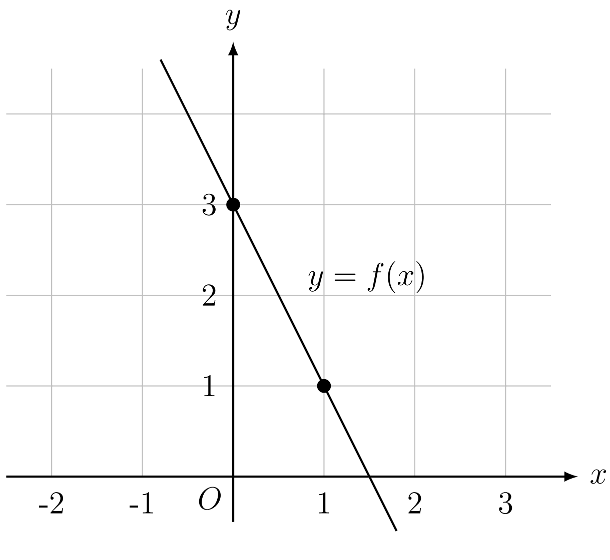
A) \$75

B) \$125

C) \$150

D) \$275

Question 31.



[Tg:@abusat_760] The graph of the linear function f is shown in the xy-plane above. The graph of the linear function h (not shown) is perpendicular to the graph of f and passes through the point $(1, 1.5)$. what is the value of $h(0)$?

Question 32.
[Tg:@abusat_760] What value of x satisfies the equation $4x + 4 = 28$?

A) 3

B) 6

C) 8

D) 32

Question 33.
[Tg:@abusat_760] If $\dfrac{2n}{7} = 6$, what is the value of $2n + 1$?

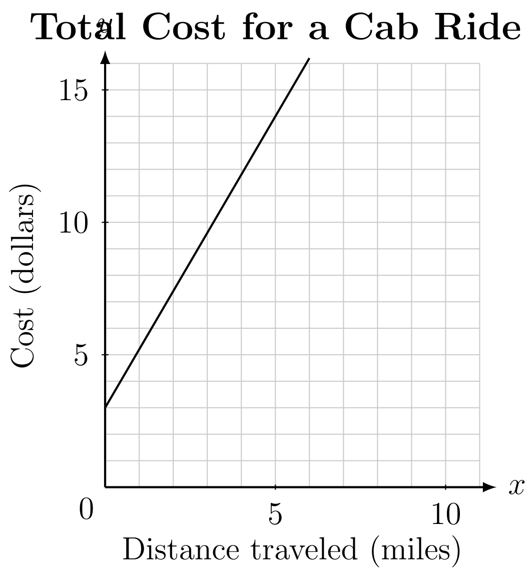
A) 22

B) 43

C) 52

D) 84

Question 34.
[Tg:@abusat_760] The line graphed in the xy-plane below models the total cost in dollars for a bike ride y, based on the number of miles traveled, x.



According to the graph, what is the cost for each additional mile traveled, in dollars, of a bike ride?

A) \$2.00

B) \$2.60

C) \$3.00

D) \$5.00

Question 35.

[Tg:@abusat_760] A company purchased a machine value at 150,000. The value of the machine depreciates by the same amount each year so that after 12 years the value will be \$30,000. Which of the following equations gives the value, v, of the machine, in dollars, t, years after it was purchased for $0 \leq t \leq 12$?

A) $v = 30,000 - 10,000t$

B) $v = 150,000 - 10,000t$

C) $v = 150,000 + 10,000t$

D) $v = 150,000 - 30,000t$

Question 36.

[Tg:@abusat_760] Line m in the xy-plane contains the point $(2, 6)$ and $(0, 3)$. Which of the following is an equation of line m?

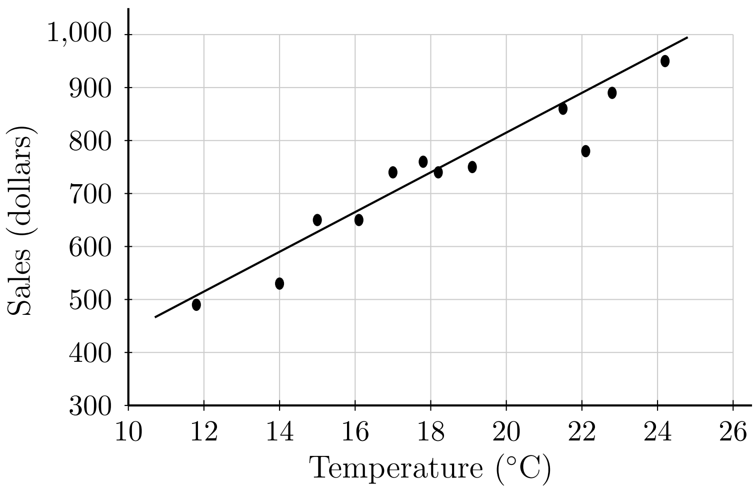
A) $y = 2x + 3$

B) $y = 2x + 47$

C) $y = \frac{3}{2}x + 3$

D) $y = \frac{3}{2}x + 1$

Question 37.



[Tg:@abusat_760] The scatterplot above shows a company's ice cream sales d, in dollars, and the high temperature t, in degrees Celsius (°C), on 12 different days. A line of best fit for the data is also shown. Which of the following could be an equation of the line of best fit?

A) $d = 0.04t + 340$

B) $d = 12t + 400$

C) $d = 28t + 340$

D) $d = 28t + 180$

Question 38.
[Tg:@abusat_760] In the xy-plane, line l has a y-intercept of -11 and is perpendicular to the line with equation $y = -\frac{2}{5}x$. if the point $(8, c)$ is on line l, what is the value of c?

Question 39.

$$20s - 12t = 6640$$

[Tg:@abusat_760] The equation above models the relationship between the speed of sound s, in meters per second, in dry air, and the temperature of the air t, in degrees Celsius. Which of the following expresses s in terms of t?

A) $s = \frac{3}{5}t + 332$

B) $s = \frac{5}{3}t + 552.5$

C) $t = \frac{3}{5}s + 332$

D) $t = \frac{5}{3}s + 552.5$

Question 40.
[Tg:@abusat_760] The equation $3m+8n = c$ can be used to determine the total charges, c, in dollars, for an order of sandwiches and salads from Mike's Catering. If m represents the number of sandwiches in the order and n represents the number of salads in the order, which of the following is the best interpretation of the number 3 in this context?

A) The number of salads in the order

B) The number of sandwiches in the order

C) The charge, in dollars, for each salad

D) The charge, in dollars, for each sandwich

Question 41.

$$-5(1-x) = 5x + b$$

[Tg:@abusat_760] In the equation above, b is a constant. If the equation has no solution, which of the following must be true?

A) $b \neq -5$

B) $b \neq -1$

C) $b \neq 1$

D) $b \neq 5$

Question 42.
[Tg:@abusat_760] In the xy-plane, line K passes through the points $(0, -3)$ and $(-2, -7)$. Line M is perpendicular to line K and has a y-intercept that is 5 units greater than the y-intercept of line K. What is an equation of line M?

A) $y = -\frac{1}{2}x - 2$

B) $y = -\frac{1}{2}x + 2$

C) $y = 2x - 2$

D) $y = 2x + 2$

Question 43.
[Tg:@abusat_760] For a service visit, a technician charges a \$26 fee plus an additional \$12 for every $\frac{1}{5}$ hour of work. If the technician's total charge was \$206, for how much time, in hours, did the technician charge?

Question 44.

$$D = \frac{5}{6}(212 - F)$$

[Tg:@abusat_760] The equation above shows how a temperature F, measured in degrees Fahrenheit, relates to the temperature D, measured in degrees Delisle. Based on the equation, which of the following must be true?

I. A temperature increases of $\frac{5}{6}$ degree Fahrenheit is equivalent to a temperature decrease of 1-degree Delisle.

II. A temperature increase of 1 degree Delisle is equivalent to a temperature decrease of 1.2 degrees Fahrenheit.

III. A temperature increases of 1-degree Fahrenheit is equivalent to a temperature decrease of $\frac{5}{6}$ degree Delisle.

A) II only

B) III only

C) II and III only

D) I, II and III

Question 45.
[**Tg:@abusat_760**] Which of the following is the equation of the line that passes through the points $(-5, -2)$, $(3, -1)$?

A) $y = \frac{1}{8}x + \frac{11}{8}$

B) $y = \frac{1}{8}x - \frac{11}{8}$

C) $y = -\frac{1}{8}x - \frac{11}{8}$

D) $y = -\frac{1}{8}x + \frac{11}{8}$

Question 46.
[**Tg:@abusat_760**] Imam opened a checking account with an initial deposit of P dollars, and then he deposited \$65 into the account each week for 15 weeks. At the end of the 15 weeks, he had deposited \$1200 into the account. If no other deposits or withdrawals were made, what is the value of P?

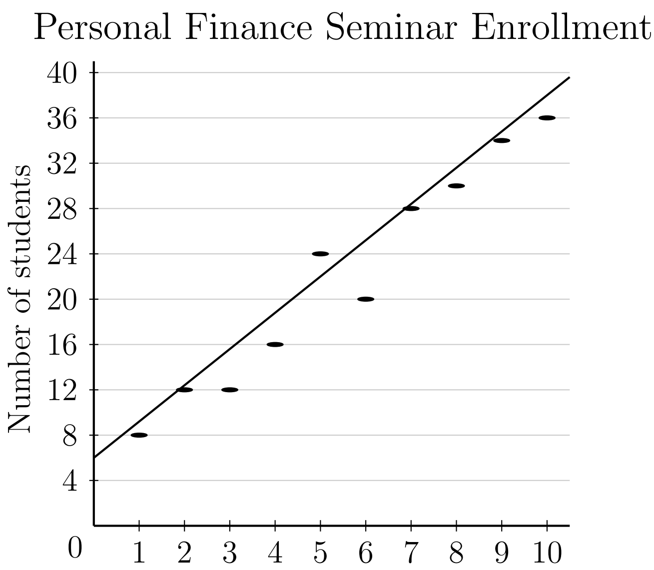
A) 65

B) 80

C) 225

D) 975

Question 47.



[Tg:@abusat_760] The number of students enrolled in a personal finance seminar for each of 10 years is shown in the scatterplot above. A line of best fit is also drawn. For how many of these 10 years was enrollment in the seminar more than predicted by this line of best fit?

A) 1

B) 4

C) 5

D) 6

Question 48.
[Tg:@abusat_760] Which of the following equations represents the line with the same x-intercept as the line with the equation $y = 2x - 5$?

A) $y = \frac{1}{3}x + 10$

B) $y = \frac{1}{3}x + \frac{10}{3}$

C) $y = -\frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + \frac{5}{3}$

Question 49.
[Tg:@abusat_760] A clown at an amusement park makes animals from balloons. She sells each animal based on the number of balloons it requires, according to the following chart:

Number of Balloons	1	2	3	4	5
Price	\$4.00	\$4.50	\$5.00	\$5.50	\$6.00

What is the price, in dollars, of an animal that takes x balloons to make?

A) x

B) $x + 4$

C) $0.5x + 4$

D) $0.5x + 3.5$

Question 50.
[**Tg:@abusat_760**] The decibel level d of each successive measure of the refrain of a song can be modeled by the equation $d = 7n + 30$, where n is the number of measures from the beginning of the refrain and $0 \leq n \leq 10$. According to the model, what is the change in decibel level between each successive measure of the refrain?

A) 7

B) 10

C) 30

D) 37

Answer Sheet

1	2	3	4	5	6	7	8	9	10
B	C	A	A	D	D	B	C	B	25

11	12	13	14	15	16	17	18	19	20
D	D	2000	A	C	C	B	C	C	D

21	22	23	24	25	26	27	28	29	30
A	D	D	C	10	D	A	A	A	A

31	32	33	34	35	36	37	38	39	40
1	B	B	A	B	C	C	9	A	D

41	42	43	44	45	46	47	48	49	50
D	B	3	C	B	C	C	D	D	A